

# ROHIT RAVINDRA POKLE

Embedded Software Engineer | Firmware Developer | OPC UA | Industrial Automation | Qt | CAN | Medical Devices | IoT

Designing scalable embedded firmware and industrial automation solutions using BLE, Matter, OPC UA, ESP32, Qt, and modern embedded systems.

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## PROFILE SUMMARY

- Firmware & Embedded Software Engineer with 3+ years of experience** designing, developing, and optimizing embedded software for IoT, medical devices, and industrial systems, delivering robust, scalable, high-performance solutions across multiple product lines.
- Hands-on with OPC UA and industrial protocols**, integrating CNC and CMM machine data using OPC UA over Ethernet for real-time quality/tolerance monitoring, machine health tracking, and production reporting in precision manufacturing environments.
- Skilled in implementing and optimizing BLE, BLE Mesh, Matter, UDP, and MQTT protocols**, enabling secure, low-latency, interoperable device communication across mobile, cloud, and embedded platforms.
- Deep expertise in C, C++, Embedded C, Python, and ESP-IDF**, developing production-grade firmware for ESP32-C3 and ESP32-S3 platforms.
- Experienced in developing Qt (C++) and Python applications** for medical devices and industrial automation systems, including a patient monitoring GUI and PC-based M2M automation for industrial test lines.
- Resourceful in full lifecycle firmware development**, including driver integration and hardware-software interfacing, ensuring reliable device operation and seamless OTA firmware upgrades.
- Proficient in designing automation scripts, log analyzers, and debugging tools**, reducing validation time by 20-25% and enhancing QA testing accuracy.

## CORE COMPETENCIES

- Embedded Software Development (C, Embedded C, C++)
- OPC UA & Industrial Protocol Integration (CNC, CMM)
- Qt + Python Application Development (Medical & Industrial GUI)
- Python
- Protocol Implementation (BLE, Matter, UDP, MQTT)
- CAN Protocol & Embedded Communication
- PC-Based Automation & PLC/M2M Integration
- Driver Development & Device Control
- Firmware Development & Debugging
- Cross-Functional Team Collaboration

## TECHNICAL SKILLS

|                           |                                                                                                                                                 |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Programming Languages     | C, C++, Embedded C, Python                                                                                                                      |
| Software Practices        | RTOS (FreeRTOS), Embedded Linux, Device Drivers, Multithreading, OOP, Design Patterns, Unit Testing, CI/CD                                      |
| Development Tools         | Qt (Qt Creator, Qt Widgets), ESP-IDF, VS Code, Arduino IDE, Oracle VM                                                                           |
| Protocols & Communication | OPC UA (Client), CAN Bus, CANopen, UART, SPI, I2C, Serial Communication, BLE, BLE Mesh, Matter, TCP/IP, Socket Programming, UDP, MQTT, Ethernet |
| Industrial & Automation   | CNC & CMM Machine Data Integration, Industrial Automation, M2M Automation, Ethernet-based Data Acquisition                                      |
| Debugging & Testing Tools | Wireshark, CAN Analyzer Tools, Serial Monitor (Tera Term, MobaXterm), Charles Proxy                                                             |
| Version Control           | Git                                                                                                                                             |
| Operating Systems         | Linux (Ubuntu), Windows, macOS                                                                                                                  |
| AI & Productivity Tools   | ChatGPT, TensorFlow Lite, Matplotlib                                                                                                            |
| Hardware Tools            | Oscilloscope, Multimeter, Soldering Station, Logic Analyzer, X-ray Optical Inspection Systems                                                   |

## DOMAIN EXPOSURE

Embedded Systems | Medical Device Software | Industrial Automation | Electronic Warfare & Avionics | IoT | Consumer Electronics | Smart Home Devices | Electric Vehicles

## EDUCATION

- 2021 - 2023** | The Oxford College of Science, Bengaluru, India | Master's in Electronics | CGPA: 7.8
- 2018 - 2021** | MES M.M. Arts and Science College, Sirsi, India | Bachelor's in Electronics | 76.6%
- 2016 - 2018** | Y.T.S.S, Yellapur, India | Pre-University (PCMB) | 80.76%

## WORK EXPERIENCE

### Oct 2025 - Present | Bharat Electronics Limited, Bengaluru, India | Embedded Software Engineer

Pay role: Numeric Technology Pvt. Ltd.

- **Qt-Based Medical Device GUI Development:** Designing and enhancing Qt-based GUI applications for medical equipment with focus on usability, stability, and performance in real-time operating environments.
- **Patient Monitoring System Development:** Developed a Qt and Python-based patient monitoring GUI integrating 5 vital-sign sensors (ECG, SpO2, temperature, blood pressure, pulse), where Python was responsible for sensor data acquisition and processing, and Qt delivered the real-time graphical user interface (GUI).
- **OPC UA Integration & Industrial Data Monitoring:** Implemented OPC UA-based communication with 3 CNC machines (two 5-axis, one 3-axis CNC machine, Siemens 840D sl controllers) and CMM systems over Ethernet for quality and tolerance monitoring, machine health tracking, and production reporting.
- **PCB Test Line Automation (Electronic Warfare & Avionics):** Developed PC-based automation using Python and industrial protocols over Ethernet for an M2M-driven PCB test line spanning soldering and X-ray Optical Inspection stations, validating board measurements against reference design data.
- **CAN Protocol & Embedded Development:** Implemented and debugged CAN protocol communication between control units and peripheral modules; wrote and optimized C/C++ embedded code for hardware interfacing and real-time data processing; performed end-to-end system debugging and validation for production-ready software.
- **Cross-Functional Collaboration:** Collaborated with hardware, testing, and systems engineering teams to ensure successful, compliant software releases, and supported Java-based configuration and diagnostic tools for system monitoring.

### Aug 2023 - Oct 2025 | Ayla Networks India Pvt. Ltd., Bengaluru, India | Firmware Engineer

Pay role: Cloud Collab Technology Pvt. Ltd.

- **Embedded Firmware Development & Ownership:** Engineered and maintained firmware for 10+ smart home devices - LED Strip Controllers, humidifiers, and smart plugs - using ESP-IDF and C.
- **Protocol Integration & Interoperability:** Integrated BLE Mesh and Matter protocols across 2 smart lighting device types (Matter-enabled smart dimmers and smart bulbs), enabling seamless connectivity and cross-device interoperability.
- **Secure OTA Firmware Deployment:** Designed and implemented OTA firmware update solutions for ESP32S3 devices, supporting secure, efficient remote upgrades and minimizing operational downtime.
- **Automation & Validation:** Developed Python-based automation scripts and log analyzers to visualize data, accelerating validation cycles and QA support.
- **Cross-Functional Collaboration & Knowledge Transfer:** Partnered with cloud and mobile app teams for end-to-end integration, and created debugging tools and training modules for QA.

### Sep 2022 - Nov 2022 | Palis Eco Vehicle, Bengaluru, India | Embedded Trainee Engineer

- **Microcontroller & Motor Driver Expertise:** Gained hands-on exposure to microcontroller interfacing, motor drivers, and power optimization strategies for electric vehicle systems.
- **Embedded Code Development & Hardware Validation:** Contributed to embedded software debugging and hardware validation for in-house vehicle control units, identifying and resolving functional discrepancies.

## PROJECTS UNDERTAKEN

### Patient Monitoring System

Role: Firmware/Software Engineer Technologies: Qt, Python, Sensor Interfacing

Description: Built a Qt and Python patient monitoring GUI integrating 5 vital-sign sensors (ECG, SpO2, temperature, BP, pulse), with Python handling data acquisition and Qt delivering the GUI.

### OPC UA Machine Data Integration

Role: Embedded Software Engineer Technologies: OPC UA, Ethernet, CNC, CMM

Description: Implemented OPC UA communication with CNC and CMM machines over Ethernet for real-time quality/tolerance monitoring and production reporting.

### PCB Test Line Automation

Role: Automation Engineer Technologies: Python, Industrial Protocols, Ethernet, X-ray Optical Inspection

Description: Automated an M2M-driven PCB test line (soldering + X-ray inspection) for electronic warfare/avionics boards, validating measurements against reference design data.

### Medical Device Qt Application

Role: Embedded Software Engineer Technologies: Qt (C++), CAN Protocol, Embedded C

Description: Designed Qt-based UI screens for medical equipment with real-time data updates from embedded controllers via CAN communication.

### Smart Lighting Control System

Role: Firmware Developer Technologies: ESP32H2, BLE Mesh, OTA Updates

Description: Built embedded firmware for BLE Mesh-based light fixtures with OTA updates and real-time command/status exchange.

### CTC Humidifier

Description: Designed a UDP-based firmware communication module for low-latency cloud control, stable across Android, iOS, and Linux.